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GOVERNEMENT OF KARNATAKA
KARNATAKA SCHOOL EXAMINATION AND ASSEMENT BOARD
6th CROSS, MALLESHWARAM, BENGALURU-560003

2026-27 II PUC MODEL QUESTION PAPER-1

Subject: 34 - Chemistry | **Maximum Marks:** 70
Time: 3.00 Hours | **Number of Questions:** 46

Instructions

1. Question paper has FIVE parts. All parts are compulsory.
2. a. Part-A carries 20 marks. Each question carries 1 mark.
   b. Part-B carries 06 marks. Each question carries 2 marks.
   c. Part-C carries 15 marks. Each question carries 3 marks.
   d. Part-D carries 20marks. Each question carries 5 marks.
   e. Part-E carries 09 marks. Each question carries 3 marks.
3. In Part-A questions, **first attempted answer** will be considered for awarding marks.
4. Write balanced chemical equations and draw neat labeled diagrams and graphs wherever necessary.
5. Direct answers to the numerical problems without detailed steps and specific unit for final answer will not carry any marks.
6. Use log tables and simple calculator if necessary (use of scientific calculator is not allowed).
7. For a question having circuit diagram/figure/ graph/ diagram, alternate questions are given at the end of question paper in a separate section for visually challenged students.

PART-A

I. Select the correct option from the given choices. **$15 \times 1 = 15$**

1. The mass percentage(W/W) of glucose in water is 10% means.
   (a) 10g of glucose dissolved in 100 g of water (b) 10g of glucose dissolved in 90 mL of water
   (c) 10g of glucose dissolved in 100 mL of water (d) 10g of glucose dissolved in 90 g of water
2. Which of the following elements is liquid at normal temperature?
   (a) Zinc (b) Mercury (c) Aluminum (d) Water
3. The reagent used in Sandmeyer's reaction is
   (a) N_2 gas (b) Cu and HCl (c) Cu_2Cl_2 and HCl (d) $CuCl_2$
4. The main natural source of acetic acid is
   (a) milk (b) vinegar (c) red ant (d) butter
5. The reaction of aniline, which produces Zwitter ion as a major product is
   (a) direct nitration (b) bromination (c) sulphonation (d) Friedel-craft alkylation
6. The method used to separation of isomeric products obtained when phenol reacts with dil. HNO_3 at low temperature is
   (a) steam distillation (b) sublimation (c) electrolysis (d) solidification
7. The ligand which forms more stable coordination complexes is
   (a) Cl^- (b) $C_2O_4^{2-}$ (c) Br^- (d) H_2O
8. Which of the following groups when present at para position increases the basic strength of aniline?
   (a) $-NO_2$ (b) -Br (c) $-NH_2$ (d) -COOH

9. In the following acids, vitamin is

(a) ascorbic acid (b) adipic acid (c) aspartic acid (d) saccharic acid

10. The best reagent useful for separation and purification of aldehyde from ketone is

(a) Tollens' reagent (b) sodium hydrogen sulphite

(c) sodium sulphate (d) 2,4-DNP reagent

11. **Statement–1:** If on mixing the two liquids, the solution becomes hot, it implies that it shows negative deviation from Raoult's law.

Statement–2: Solutions which show negative deviation are accompanied by decrease in volume.

(a) Both Statement–1 and Statement–2 are true. (b) Statement–1 is true but Statement–2 is false;

(c) Both Statement–1 and Statement–2 are false. (d) Statement–1 is false but Statement–2 is true.

12. Match the following transition metal/compounds with their catalytic activity in the corresponding processes.

Transition metal/compounds	Name of the process
(i) $TiCl_4 + Al(CH_3)_3$	(A) Wacker process
(ii) $PdCl_2$	(B) Contact process
(iii) Ni	(C) manufacture of polyethene
(iv) V_2O_5	(D) hydrogenation of fat

(a) i-C, ii-A, iii-D, iv-B (b) i-B, ii-A, iii-D, iv-A (c) i-A, ii-C, iii-D, iv-B (d) i-D, ii-A, iii-B, iv-C

13. Cumene hydroperoxide on hydrolysis with dilute acids gives

(a) phenol and oxygen (b) phenol and hydrogen

(c) hydrogen and oxygen (d) phenol and acetone

14. Among the following cells, the cell used in the apollo space program for providing electric power is

(a) SHE (b) H_2-O_2 fuel cell (c) Daniel cell (d) Mercury cell

15. The following results have been obtained during kinetic studies of the reaction $A_{(aq)} + 2B_{(aq)} \longrightarrow C_{(aq)}$.

Choose the correct option for the rate equation for the above reaction.

Experiment	Concentration of [A] $(molL^{-1})$	Concentration of [B] $(molL^{-1})$	Rate of formation of 'C' $(molL^{-1}\,min^{-1})$
I	0.1	0.1	6.0×10^{-3}
II	0.2	0.3	7.2×10^{-2}
III	0.1	0.4	2.4×10^{-2}
IV	0.4	0.3	2.88×10^{-1}

(a) $Rate = K[A]^1[B]^2$ (b) $Rate = K[A]^1[B]^0$

(c) $Rate = K[A]^2[B]^1$ (d) $Rate = K[A]^4[B]^1$

II. Fill in the blanks by choosing the appropriate word from those given in the brackets:

(two, EDTA, chloroform, three, nucleotide, one) **5 ×1 = 5**

16. The polyhalo compound used as an aesthetic during surgery was _____.

17. Nucleic acids are the long chain polymers of _________.

18. Van't Hoff factor for KCl solution assuming the complete dissociation is_______.

19. The number of hydroxyl groups present in glycerol is __________.

20. Example for polydentate ligand is ________.

PART – B

III. Answer any three of the following. Each question carries 2 marks. **3 × 2 = 6**

21. What is Lucas reagent? Which class of alcohol does produces turbidity immediately with it?

22. Write the S_N2 mechanism for conversion of chloromethane to methanol.

23. For the reaction: $H_2(g) + I_2(g) \longrightarrow 2HI\ (g)$; draw the diagram showing plot of potential energy versus reaction coordinate to explain the role of activated complex in a reaction.

24. Name the hormone released rapidly due to rise in blood glucose level to keep the blood glucose level within the narrow limit. Mention the number of amino acids present in this hormone.

25. What are transition elements? Give an example.

PART – C

IV. Answer any three of the following. Each question carries 3 marks. **3 × 3 = 9**

26. Using Valence Bond Theory [VBT] explain hybridisation, geometry and magnetic property of $[CoF_6]^{3-}$ ion. [Given: Atomic number of cobalt is 27]

27. Heteroleptic complexes with co-ordination number 6 show geometrical isomerism. A complex $[MA_3B_3]$ shows geometrical isomerism. If central metal ion M has +3 oxidation state. Predict the denticity and draw the structure of two geometrical isomers of the complex.

28. What is Lanthanoid contraction? Mention two consequence of it.

29. How many square pyramidal units present in decacarbonyldimanganese(0)? Write its molecular and structural formulae.

30. Complete the following equations.

(a) ________ $+\ 8\ Na_2CO_3 +\ 7\ O_2 \longrightarrow 8\ Na_2CrO_4 +\ 2\ Fe_2O_3 +\ 8\ CO_2$

(b) $MnO_2 \xrightarrow[\text{Oxidised with air}]{\text{Fused with KOH}}$ ________

(c) $2\,KMnO_4 \xrightarrow{513K} K_2MnO_4\ +\ O_2\ +$ _______

V. Answer any two of the following. Each question carries 3 marks. **2 × 3 = 6**

31. Mention three factors that affect the rate of a chemical reaction.

32. Name the concentration term which is commonly used in medicine and pharmacy. Write the definition and mathematical equation for that concentration term.

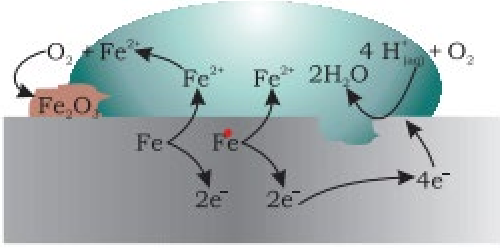
33. Depending on the magnitude of conductivity of the materials, mention the three types of materials.

34. The diagram shows that when iron is exposed to atmospheric air,

a) Name the phenomenon involved in this diagram.

b) write the atmospheric oxidation reaction of iron.

c) mention any one method to avoid this phenomenon.



PART – D

VI. Answer any four of the following. Each question carries 5 marks. **4 × 5 = 20**

35. (a) Write the Haworth structure of maltose. Why they show reducing property?

(b) What are fibrous proteins? Name the protein present in hair. **(3+2)**

36. (a) Write the three steps involved in the acid catalysed dehydration of ethanol to ethoxy ethane at 413 K.

(b) Explain Kolbe's reaction with equation. **(3+2)**

37. (a) When alkyl chlorides reacts with sodium iodide in dry acetone gives alkyl iodide.

(i) Name this reaction. (ii) Write the general equation. (iii) Mention the role of dry acetone.

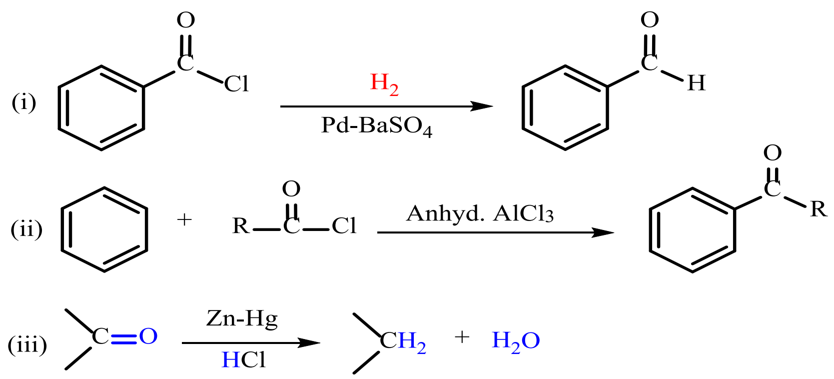
(b) What is meant by racemic modification? "They are optically inactive". Give reason. **(3+2)**

38. (a) Which class of amines undergoes carbylamine reaction (isocyanide test)? Explain this reaction with a general chemical equation.

(b) Identify the compounds "X" and "Y" in the following conversion.

$$C_6H_5\text{–}NH_2 \xrightarrow[273-278K]{NaNO_2 + 2HCl} \text{"X"} \xrightarrow{\text{"Y"}} C_6H_5\text{–}I + KCl + N_2$$ **(3+2)**

39. (a) Mention the name of the following reactions.



(b) Among methanal and ethanal, which one undergoes Cannizaro reaction? Give reason. **(3+2)**

40. The compound "A" has molecular formula C_7H_8 is heated with alkaline $KMnO_4$ followed by acidification gives compound "B" of the molecular formula $C_7H_6O_2$. This compound "B" turns blue litmus to red. The sodium salt of compound "B" is heated with reagent "X" gives hydrocarbon "C" of molecular mass 78 g mol^{-1}.

(a) Write the structure of compounds "A", "B" and "C".

(b) Identify "X" and mention its role in the reaction. **(3+2)**

PART – E (PROBLEMS)

VII. Answer any three of the following. Each question carries 3 marks. **3 × 3 = 9**

41. The cell in which the reaction occurs $2Fe^{3+}_{(aq)} + 2I^-_{(aq)} \rightleftharpoons 2Fe^{2+}_{(aq)} + I_{2(s)}$; $E^o_{cell} = 0.236\,V$ at 298 K

Calculate the value of log K_c (K_c = equilibrium constant) of the cell reaction.

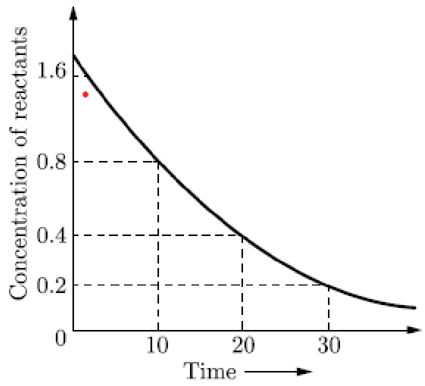
42. The first order rate constant at 600 K is $2\times10^{-5}\,s^{-1}$ and energy of activation is 209.8 $kJ\,mol^{-1}$ for a reaction. Calculate the rate constant at 700 K. [Given: $R=8.314\,JK^{-1}mol^{-1}$]

43. Henry's constant for the molality of methane in benzene at 298 K is 4.27×10^5 mm Hg. Calculate the mole fraction of methane in benzene at 298 K under 760 mm Hg.

44. Calculate the limiting molar conductivity of Cl^- ion by using the data: $\lambda_{Ca^{2+}} = 119.0\,S\,cm^2mol^{-1}$ and Λ^o_m for $CaCl_2 = 271.6\ S\,cm^2mol^{-1}$.

45. Calculate the osmatic pressure in pascal exerted by a solution prepared by dissolving 0.925 g of polymer of molar mass 1,85,000 in 500 mL of water at $37^{o}C$. [Given: $R = 8.314\times10^{3}\ Pa\,L\,K^{-1}\,mol^{-1}$]

46. Analyse the given graph, drawn between concentration of reactant in $molL^{-1}$ v/s time in minute, calculate the average rate of the reaction in terms of minutes and seconds.



PART-F

(For visually challenged students only)

34. Define the term corrosion. Suggest two methods to prevent corrosion of metal.

46. At 318 K, for the reaction $2N_2O_{5\,(g)} \longrightarrow 4NO_{(g)} + O_{2\,(g)}$ in CCl_4, the initial concentration of N_2O_5 is $2.33\,mol\,L^{-1}$ in reduced to $2.08\,mol\,L^{-1}$ in 100 minutes. Calculate the average rate of reaction in terms of seconds.

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